Suyash Vardhan Singh

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Education

University of South Carolina — Columbia, SC
Ph.D. in Computer Engineering, **GPA: 3.60** — Jan 2022 – Dec 2026
Dissertation: Hardware–Software Co-Design and Design-Space Exploration of Processor Architectures for Efficient GEMM

California State University Fullerton — Fullerton, CA
M.S. in Computer Engineering — May 2019

IET Lucknow — Lucknow, India
B.Tech. in Electronics and Communication Engineering — Jun 2017

Research Interests

Computer Architecture, High-Performance Computing, AI Accelerator Systems, On-Chip Interconnects and Data Movement, Distributed and Systolic Computation, Hardware/Software Co-Design, ML-Driven System Optimization, Resource Scheduling, Performance Modeling, Design-Space Exploration, Reconfigurable Computing, and Emerging Computing Architectures.

Research and Engineering Experience

University of South Carolina — Columbia, SC
Research Assistant / Teaching Assistant — Jan 2022 – Present

- Architected **RISCA**, a configurable many-core VLIW RISC-V accelerator that integrates vector processing with a two-dimensional systolic interconnect for scalable, throughput-oriented AI and GEMM workloads.
- Co-designed RISCA's processor array, distributed scratchpads, streaming data paths, double-buffered memory hierarchy, and compiler support to coordinate computation and communication across processing elements.
- Developed custom **LLVM** backend support for instruction scheduling, VLIW packetization, software pipelining, hazard management, and communication-aware kernel execution on the accelerator.
- Built RTL prototypes and cycle-level simulation workflows to evaluate architectural alternatives using latency, throughput, memory bandwidth, FPGA resource utilization, clock frequency, and scalability metrics.
- Developed **N-TORC**, an ML-driven design-space exploration and resource-scheduling framework for FPGA-based scientific and real-time AI workloads, enabling performance prediction and automated selection of hardware configurations under execution and resource constraints.
- Created analytical and machine-learning models for system latency, throughput, resource utilization, and scheduling behavior, and validated predictions using simulation and FPGA implementation results.
- Investigated communication, computation, and memory-bandwidth bottlenecks in accelerator systems, including tiled data movement, streaming execution, workload mapping, and synchronization across processing elements.
- Developed research artifacts and hardware prototypes using SystemVerilog, Verilog, C/C++, Python, LLVM, HLS, gem5, Spike, and Xilinx FPGA platforms.
- Published first-author research at **FCCM, FPGA, and GLSVLSI** and contributed to collaborations spanning computer architecture, machine learning systems, scientific computing, and emerging hardware.
- Served as a teaching assistant for CSCE 611 (Advanced Digital Design), mentoring students in pipelined RISC-V processor design, computer architecture, RTL implementation, and hardware verification.

Pacific Microchip Corp. — Culver City, CA
Design Engineer Intern — Mar 2020 – Apr 2020

- Developed fixed-point signal-processing models, including FFT components, using MATLAB and Simulink.
- Automated RTL-generation and FPGA-build workflows and implemented generated hardware components for functional verification.

HDLexpress — Canby, OR
RTL Design Intern — Jul 2019 – Jan 2020

- Designed and verified a read-only instruction-cache controller in SystemVerilog.
- Debugged cache-control and memory-interface logic through simulation and structured design verification.

California State University Fullerton — Fullerton, CA
Research Assistant — Mar 2018 – May 2019

- Conducted research in low-power IC design, processor pipelines, digital circuit optimization, and FPGA-based system prototyping.
- Performed transistor-level characterization using Synopsys HSpice and implemented digital prototypes using Xilinx Vivado.

Research Service: Reviewer, **IEEE International Symposium on Circuits and Systems (ISCAS)**, 2025–2026.

Selected Publications

- Suyash V. Singh, I. Ahmad, D. Andrews, M. Huang, A. Downey, and J. Bakos, "N-TORC: Native Tensor Optimizer for Real-time Constraints," **IEEE Symposium on Field-Programmable Custom Computing Machines (FCCM)**, 2025. DOI: 10.1109/FCCM62733.2025.00061
- Suyash V. Singh, A. Kolinko, M. H. Amin, R. Zand, and J. Bakos, "A Decomposition-Based Memristive Crossbar Solver and FPGA-Accelerated Hardware Implementation," **ACM Great Lakes Symposium on VLSI (GLSVLSI)**, 2025. DOI: 10.1145/3716368.3735282
- Suyash V. Singh, I. Ahmad, D. Andrews, M. Huang, A. Downey, and J. Bakos, "Resource Scheduling for Real-Time Machine Learning," **ACM/SIGDA International Symposium on Field-Programmable Gate Arrays (FPGA)**, 2025. DOI: 10.1145/3706628.3708848
- Suyash V. Singh and R. Mahto, "Switchable Single/Dual Edge Registers for Pipeline Architecture," **ITNG**, 2019, pp. 535–541. DOI: 10.1007/978-3-030-14070-0_79

Technical Skills

Programming and Hardware Languages: C, C++, Python, Bash, Verilog, SystemVerilog, MATLAB
Architecture and Systems: RISC-V, VLIW and Vector Architectures, Systolic Arrays, On-Chip Interconnects, Memory Hierarchies, Dataflow Architectures, Resource Scheduling, Performance Modeling, Design-Space Exploration, FPGA and ASIC Design
Compiler and Simulation: LLVM, gem5, Spike, Cycle-Level Simulation, Software Pipelining, Instruction Scheduling, Analytical Modeling
AI and Acceleration: Machine-Learning-Based Performance Prediction, AI Accelerator Design, Real-Time ML Systems, GEMM Optimization, High-Level Synthesis
Tools and Platforms: Vivado, Vitis, Vitis HLS, Synopsys VCS, Design Compiler, QuestaSim, HSpice, LTSpice, Git, Linux, AMD/Xilinx FPGA Platforms